

Secondary Treatments in powder metallurgy

Why secondary operations are required in powder metallurgy products?

Surface smoothness and finish: Sintered parts may have rough surfaces. Machining, grinding, or shot peening can improve surface finish and enhance aesthetics

Dimensional Accuracy: Sintering can lead to dimensional variations, so secondary operations like sizing and repressing can improve tolerances and achieve tighter dimensional control.

Performance Enhancement: Secondary operations like heat treating, plating, and infiltration can significantly impact the mechanical properties, corrosion resistance, or other performance characteristics of the final part.

Functional Requirements: Some applications require specific features like threads, intricate shapes, or surface coatings that can be achieved through secondary operations like machining, plating, or coating.

Cost Reduction: In some cases, secondary operations can reduce the overall cost of manufacturing by minimizing the need for complete machining of the part.

Secondary treatments:

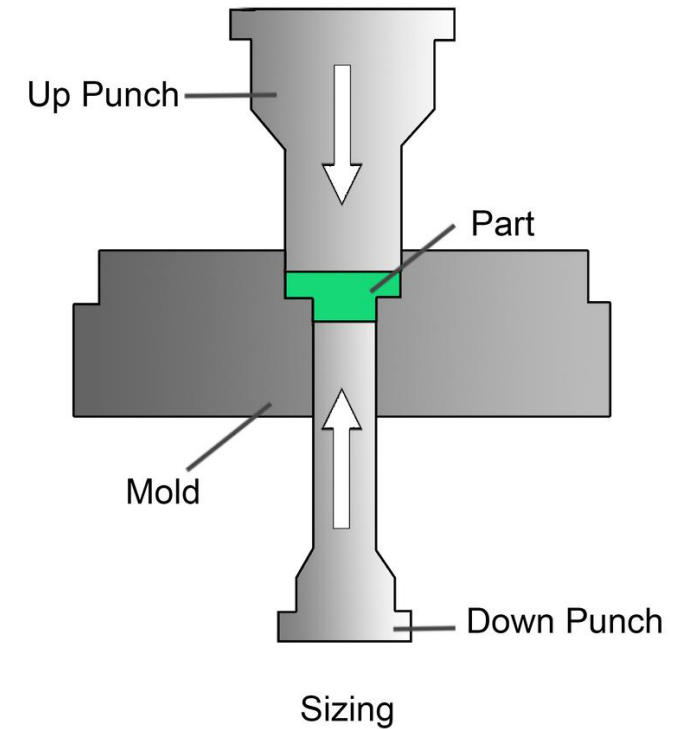
- Sizing and coining
- Machining
- Impregnation
- Infiltration
- Surface engineering
- Heat treatment
- Joining

Sizing:

- Sizing is done to refine dimensional accuracy.
- Pressures used are no more than the initial compacting pressure.
- Sized parts will be straighter, dimensional tolerances will be closer and surface finish will be improved.
- Little or no increase in density is achieved.

Coining:

- Pressures used are more than the initial compacting pressure.
- Increased part density in addition to improving dimensional accuracy.
- Increased mechanical properties.



Machining:

- Machining is any of various processes in which a piece of raw material is cut into a desired final shape and size by a controlled material removal process.
- Machining creates geometric features that cannot be achieved by pressing, such as threads, side holes, and other details

Impregnation:

- Controlled porosity permits PM parts to be impregnated with oils or resins.
- This can be done simply by submerging the PM components in an oil bath for several hours.
- Vacuum impregnation process gives best results.
- In resin impregnation interconnected pores are filled with resins, usually polyesters of the thermosetting type.
- This process greatly improves the machinability of PM parts by filling the pores and increasing the density.
- Entrapment of other fluids is prevented.

Infiltration:

- Operation in which the pores of the PM part are filled with a molten metal.
- The melting point of the filler metal must be below that of the PM part.

$$T_M (\text{filler}) < T_M (\text{Part})$$

- Involves heating the filler metal in contact with the sintered component so capillary action draws the filler into the pores.
- Resulting structure is relatively nonporous, and the infiltrated part has a more uniform density, as well as improved toughness and strength.

Shot peening:

- Shot peening is a cold work process used to finish metal parts to prevent fatigue and stress corrosion failures and prolong product life for the part.
- In shot peening, small spherical shot bombards the surface of the part to be finished.
- The shot acts like a peen hammer, dimpling the surface and causing compression stresses under the dimple.
- As the media continues to strike the part, it forms multiple overlapping dimples throughout the metal surface being treated.
- Significant improvement in fatigue properties can result from the formation of harder and stronger surfaces.

Joining:

- Powder metallurgy itself makes it possible to obtain complex shapes, and hence the joining operation is not often required.
- Conventional welding methods (TIG, MIG, electron beam, resistance and friction welding) are used to join PM parts.
- Best welding results for PM steels are achieved when the component density is 6.8 g/cm³ (88% of theoretical density) or higher.
- Sinter welding can also be done by inserting suitable powder of the same metal between the parts to be joined, and then hot pressing the assembly in a suitable atmosphere.

